CHAPTER 4

Derivation: from tangent to tangent

Chapter overview

Objectives. Interpret a rate; approach a derived number; determine a tangent; understanding differentiability failures.
Prerequisites. Images, slope of a straight line, literal calculation, square root and absolute value.
Annual benchmark. 3·12 November: 8 a.m.

Diagnostic check: six questions, 10 minutes

Answer without consulting the course. Indicate the questions for which assistance is required. diagnostic check is not graded.

1. Pent of the right passing through (1; 2) and (3; 8) ?

2. Develop $(2 + \quad h)^2 - 4$.

3. Simplify $\dfrac{4h + h^2}{h}$ if $h \neq 0$.

4. Calculate $\sqrt{0}$ and $\sqrt{0{,}01}$.

5. Calculate $|-3|$ and $|2|$.

6. A slope 2 line passes through (1; 5). Equation ?

After the diagnostic check. Retrieve the concepts indicated with the teacher; repeat a question and then explain the rule used.

Speaking and reasoning. Explain to a comrade why we simplify the rate before making it tender h around 0. Distinguish medium and instant speed.

Search before formalizing

For $f(x) = x^2$, vary h positive and negative around 0 keeping the A point abscisse 1.
Draw three scissors; conjecture a boundary line and determine it without graphical reading.

Organization. Individual research 5 min, confrontation 2 5 min, pooling 5 min.
Keep a trace of a test, even if not completed. A conjecture must then be justified.

4.1 Rates of change and scales

Definition and results

For $a \neq b$ in f, the rate of change between a and b is $\dfrac{f(b) - f(a)}{b - a}$. It is the slope of the slatting through the corresponding points.
unit of f divided by that of the variable.

Status and justification. Definition; formula of the slope of a line already studied. $b = a + h$, $h \neq 0$, the rate becomes $\dfrac{[f(a+h) - f(a)]}{h}$.

Example solved. For $f(x) = x^2$, between 1 and 3, the rate is $\dfrac{9-1}{3-1}$ = 4. Between 1 and 1 + h, it's worth $2 + h$. The slopes for $h = 0{,}1$ and $h = -0{,}1$ is 2.1 and 1.9.

Point of vigilance

The rate is not $\dfrac{f(b)}{f(a)}$; h represents a gap, positive or negative depending on the field.

Immediate application. Calculate the rate of $f(x) = x^2 + 3x$ between 2 and 2 + h.

4.2 Derivative number

Definition and results

f is derivable if the rate $\dfrac{[f(a+h) - f(a)]}{h}$ tends to a real when h tends to 0, in the field. $f\prime(a)$ An intuitive approach to the boundary is used here.
kinematics, if f gives a position in meters and time in seconds, the derived number means an instantaneous speed in m s $^{-1}$.

Status and justification. For $f(x) = x^2$, the rate is 2 $a + h$, close to 2 a ; so $f\prime(a) = 2a$. This calculation will be taken as derivative function in Chapter 6.

Example solved. For $s(t) = t^2 + 3t$, the rate around 2 is 7 + h, therefore $s\prime(2) = 7\ \mathrm{m\,s^{-1}}$. The average speed on [2; 3] is 8 m s $^{-1}$; it is not speed at moment 2.

Point of vigilance

We do not replace h by 0 in a non-simplified quotient: we would get a division by 0.

Immediate application. For $f(x) = 3x^2 - 2x$, calculate $f\prime(1)$ from the rate.

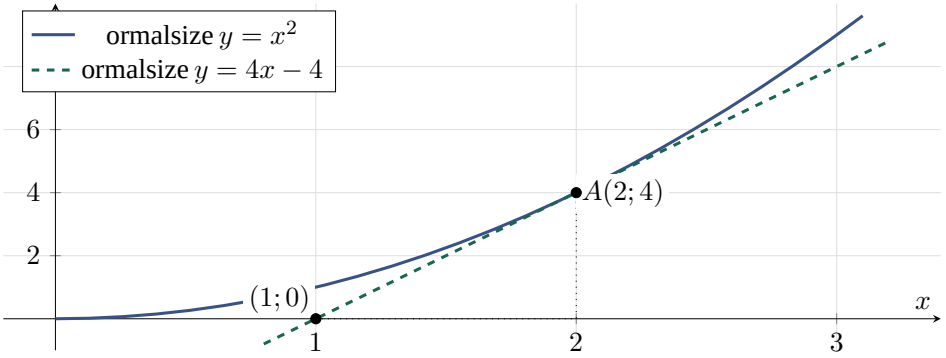
4.3 Tangent and Affine Approximation

Definition and results

If f is derivable from a, the tangent at point ($a; f(a)$ a for equation $y = f(a) + f\prime(a)(x - a)$. For h small, we use $f(a + h) \approx f(a) + f\prime(a)h$. This local approximation does not provide it alone is a bound of error.

Status and justification. The tangent has sloped $f\prime(a)$ and passes through $(a; f(a))$. $y = f\prime(a)x + p$, substitute this point $p = f(a) - af\prime(a)$, hence the equation.

Example solved. For $f(x) = x^2$ in 2, tangent $y = 4 + 4(x - 2) = 4x - 4$. Approach 2 ,$01^2$ per 4 + 4(0 ,$01) = 4$,04; the exact value is 4 ,0401, deviation $0{,}0001 = h^2$.



The A tangent also passes through (1; 0). Its slope is $\frac{4-0}{2-1}$ = 4. The discontinuous path allows the distinguish from the curve, even in black and white.

Point of vigilance

A tangent can cross the curve elsewhere. "One common point" is not a definition.

Immediate application. We give $f(3)$ = 5 and $f\prime(3) = -2$. Write the tangent and approach $f(3{,}04)$.

4.4 When Differentiability Fails

Definition and results

The square root function is defined on [0; + ∞[but is not derivable in 0. The function absolute value is defined on $\mathbb{R}$ but is not driftable in 0. Be defined or have a curve therefore does not guarantee differentiability.

Status and justification. For $h > 0$, $\dfrac{\sqrt{h} - 0}{h} = \dfrac{1}{\sqrt{h}}$, which does not tend to a real finite when h is close to 0. $|x|$, rate $\dfrac{|h|}{h}$ is 1 to the right and -1 left: no common value.

Example solved. For $|x|$, the slattings from the origin all have slope 1 to the right and -1 à left. The angular dot does not have a single finished slope tangent.
limit direction is vertical; there is no actual derivative number.

Point of vigilance

A very large slope is not proof of non-differentiability; it is necessary to explain the behaviour rate.

Immediate application. Justify that $g(x) = |x - 2|$ is not detrievable in 2.
Differentiability of $x \mapsto \overline{x}$ in 0 by its rate.

Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 4.1 · Two secantes

The function f is defined on $\mathbb{R}$ by $f(x) = x^2 - 1$. Note A the point of its curve abscisses 1.

1. Calculate the guiding coefficient of the joining splitter A at the point of the curve of abscisse 2.

2. Resume the calculation with the point of the curve of abscisse 1 ,1.

3. After study of the derived number, determine $f\prime(1)$ and compare this value to the two calculated rates. Explain what changes when the second point approaches A.

Exercise 4.2 · One tangent

The function is considered $f(x) = 2x^2 + 1$, defined on $\mathbb{R}$. The objective is to determine the equation of its tangent at the point of abscissa -1, starting from the rate of change.

1. For $h \neq 0$, write and then simplify $\dfrac{f(-1+h) - f(-1)}{h}$.

2. Deduct from it $f\prime(-1)$ Calculate also $f(-1)$.

3. Write down the reduced tangent equation and verify that it passes through the point of contact.

Fiscal year 4.3 · Marginal cost

A company uses the model $C(q) = q^2 + 20q + 100$, where $q \geq 0$ means the quantity produced and $C(q)$ The model extends the quantity by a real variable in order to study small variations in production.

1. Calculate $C(10)$ and $C(11)$, then the rate of change of C between 10 and 11. Interpret this rate in the context.
2. From the rate between 10 and 10 + h, determine $C'(10)$.
3. Compare the two results. Explain why the cost of an additional unit is not exactly equal to the derived number in this model.

Exercise 4.4 · Read a tangent

This exercise has two independent situations. A tangent to a curve is a straight: a other point of this line does not necessarily belong to the curve.

1. In the figure of the section on tangent, read the coordinates of two points on the right green. Deduce its slope and check the value announced in the legend.
2. In another situation, the tangent at the curve of a function f at point $A(2; 3)$ goes through $B(4; 2)$. Determine $f(2)$ and $f'(2)$ Is it possible to deduce from it $f(4)$?
3. Use this tangent to provide an approximate value of $f(2{,}1)$ Make the sign appear " about equal" in the conclusion.

Exercise 4.5 · Controlling an approximation

We want to approach $0{,}98^2$ without first performing the multiplication. curve of the square function at the point of abscisse 1.

1. Write the equation of this tangent and deduce the desired approximation.
2. Then calculate the exact value of $0{,}98^2$. The error is defined as "exact value minus value approach": calculate this error and specify its sign.
3. Explain the result from the identity $(1 + h)^2 = 1 + 2h + h^2$.

Exercise 4.6 · Tangente and intersection

The function f is defined on $\mathbb{R}$ by $f(x) = x^3$. It is admitted that $f'(1) = 3$. A student states that tangente and a curve always have one common point.

1. Determine the tangent equation at the curve of f at the point of abscisse 1.
2. Check by developing that $x^3 - 3x + 2 = (x - 1)^2(x + 2)$.
3. Determine all points of intersection of the curve and this tangent. Is it correct?

Exercise 4.7 · Numerical secants

To study the square function in the vicinity of 1, we define, for $h \neq 0$,

$$T(h) = \frac{(1+h)^2 - 1}{h}.$$

The six values 0 shall be used. ,1, 0,01, 0,001 and their opposites.

1. Write a Python function that calculates $T(h)$, then produce a table of six results.

2. Conjecture the value of the number derived from the square function in 1.

3. Simplify algebraically $T(h)$ to justify the conjecture. Explain why the value test
It is not enough to demonstrate the result alone.

Fiscal year 4.8 · Deferred review

The reciprocal function is defined by $f(x) = \frac{1}{x}$ for $x \neq 0$. It is studied in the vicinity of 1.

1. Specify the prohibited values of h in rate $\frac{f(1+h) - f(1)}{h}$, then simplify it.

2. Deduct from it $f'(1)$ and the tangent equation in 1.

3. Use this tangent to approach $\frac{1}{1,02}$. Maintain the distinction between exact value and value
Come here.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, Exercise 17 p. 127, all questions. Read two points of each tangent; write the
the number of increments rather than estimate its inclination.

CQFD, Exercise 19 p. 127, Questions 1 and 2. Distinguish from point of contact and
slope of the tangent right.

MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. Rate of x^2 between 1 and 2:

   A. 2
   B. 3
   C. 1
   D. 4

2. If $f(1) = 4$, $f\prime(1) = 2$, tangent:

   A. $y = 2x + 2$
   B. $y = 4x + 2$
   C. $y = 2x + 4$
   D. $y = x + 3$

3. The derived number is

   A. a function
   B. an area
   C. an abscissa
   D. a real if the derivative exists

4. For x^2 in 3, derived number:

   A. 9
   B. 3
   C. 6
   D. 12

5. The approximation of $f(a + h)$ is

   A. $f(a)f\prime(a)$
   B. $f(a) + hf\prime(a)$
   C. $f(a) + f\prime(h)$
   D. $hf(a)$

MCQ 2: understand and choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $\sqrt{x}$ in 0 is

   A. Derivable derivative 0
   B. Derivable derivatives 1
   C. Non-derivable
   D. Undefined

2. $\dfrac{|h|}{h}$ for $h < 0$ is

   A. -1
   B. 1

C. 0

D. h

3. A tangent of slope 0 is

A. Vertical

B. oblique

C. Unable

D. Horizontal

4. Position in m and time in s: derivative unit

A. m^2

B. $\mathrm{m\,s}^{-1}$

C. $\mathrm{s\,m}^{-1}$

D. m

5. For derivative a, replace $f(a+h)$ by $f(a)+hf\prime(a)$ is generally

A. an exact identity

B. a definition of h

C. Local approximation

D. Always false even for an affine

True or false?

1. A tangent can never cross a curve.

2. The absolute value function is derivatable in 0.

3. An affine function $mx+p$ a for derivative number m in all real.

Personal review

I give the domain and h not zero; I read a slope; I use the pitch point; I distinguish exact and Approached.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and a mistake to avoid.

Corrected and conduct of chapter 4

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. $\dfrac{8-2}{3-1}=3$. *Action:* Represent horizontal and vertical increments.

2. $4h+h^2$. *Action:* Develop before simplifying.

3. $4+h$. *Action:* Factorize h to the numerator before division.

4. 0 and 0,1. *Action:* Check by square elevation.

5. 3 and 2. *Action:* Represent distance at 0.

6. $y=2x+3$. *Action:* Replace the coordinates to find the original order.

Research activity

The slopes are 2 + h ; limit 2. The line passes through (1; 1), therefore $y=2x-1$. The drawing motivates, the calculation justifies.

Immediate applications

1. $f(2+h)-f(2)=7h+h^2$; rate 7 + h for $h\neq 0$.

2. The rate is 4 + 3 h, so the derivative number is 4.

3. $y=5-2(x-3)=-2x+11$; approximation 5 $-2(0{,}04)=4{,}92$.

4. The rate in 2 is $\dfrac{|h|}{h}$, with the two opposite limit values. For the root in 0, the rate $\dfrac{1}{\sqrt{h}}$ for $h>0$ exceeds any bound when h approaches 0: no actual derivative number.

Exercises: approach and results

Exercise 4.1 · Two secantes. Pentes 3 and 2.1; derived number 2. The second is closer to the tangent slope; this is an exact rate calculation.

Exercise 4.2 · A tangent. Rate 4 $a+2h=-4+2h$, therefore $f'(-1)=-4$. $f(-1)$ = 3 ; tangent $y=3-4(x+1)=-4x-1$.

Fiscal year 4.3 · Marginal cost. $C(10)=400$, $C(11)$ = 441, increment 41 for one unit. The rate around 10 is 40 + h, thus derivative 40 : approximation of the increment for a small variation, not equal for an entire unit.

Exercise 4.4 · Read a tangent. In the course figure: points (1;0) and (2;4), slope 4. the new situation, $f(2)$ = 3, slope $-\dfrac{1}{2}$. Approximation 3 $-0{,}5\times 0{,}1=2{,}95$. Ordered from B is not necessarily $f(4)$.

Exercise 4.5 · Control an approximation. $1+2(-0{,}02)=0{,}96$. Right 0 ,9604; exact less approach 0 ,0004 >0. The term omitted is h^2 : default approximation.

Exercise 4.6 · Tangente and intersection. The tangent has as an equation $y=1+3(x-1)=3x-2$. The development of the product gives well x^3-3x +2. The apses of the intersections are therefore 1 and -2, by the zero-product rule. The points are (1; 1) and (-2; $-$There are two commonalities: This is enough to refute the general statement.

Exercise 4.7 · Numerical secants. List of rates 2+ h : 2.1; 2.01; 2.001 and 1.9; 1.99; 1.999. The algebraic calculation gives the real 2. At very small hours, the floating calculation can lose precision.

Exercise 4.8 · Deferred review. For $h\neq 0,-1$, the rate is $-\dfrac{1}{1+h}$; it tends towards -1. Approximation 1 $-0{,}02=0{,}98$. The exact value $\dfrac{50}{51}$ allows for control.

CQFD, Exercise 17 p. 127, all questions. The apses of A, B and C are 0, 1 and 2. tangents give respectively slopes 2, 0 and -2. Thus $f\prime(0)=2$, $f\prime(1)=0$, $f\prime(2)=-2$. The axis units shall be read on the grid.

CQFD, Exercise 19 p. 127, Questions 1 and 2. Point A reads (1; 4), therefore $f(1)=4$. The tangent passes through (0; 1) and (1; 4): slope 3. $f\prime(1) = 3$ and tangent a for equation $y=4+3(x-1)=3x+1$.

MCQ: answers and reasons

MCQ 1.

1. B. $\dfrac{4-1}{2-1}=3$.

2. A. $4+2(x-1)$.

3. D. This is a value in one point.

4. C. Rate 6 + h.

5. B. Affine tangent function.

MCQ 2.

1. C. Rates $\dfrac{1}{\sqrt{h}}$ Not finished.

2. A. $|h|=-h$.

3. D. Constant order.

4. B. A change in position divided by a change in time is expressed in m s $^{-1}$.

5. C. Accurate equality for an affine, approximation in general.

True or false

1. False: x^3 and tangent 3 $x-2$ in 1 also meet in -2.

2. False: Rates on both sides are valid -1 and 1.

3. True: the rate is exactly m for any non-zero h.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.